GHS / Weil descent on IPsec Oakley groups 3 and 4

Generalized GHS / Weil descent on the IPsec Oakley EC groups 3 and 4

Date 2026-10-23. Status: structural screen, exact magic-number spectrum, isogeny-neighbourhood census (both twists), exponent-level cost model. **No attack, no speedup, no vulnerability claim.** Nothing here is a leaderboard or benchmark row, so no `VALIDATION.json`.

Curves, checked against RFC 2409 sections 6.3 and 6.4 on 2026-10-23 (a = 0 for both):

group	field	b	RFC group order
3	`u^155 + u^62 + 1`	0x07338f	0x080000000000000000057db5698537193aef944
4	`u^185 + u^69 + 1`	0x1ee9	0x01ffffffffffffffffffffdbf2f889b73e484175f94ebc

Reproduce

```
cargo run --release --bin ghs_screen -- --degree 155 --modulus
0x800000000000000000000004000000000000001 --a 0 --b 0x07338f
cargo run --release --bin ghs_screen -- --degree 185 --modulus
0x2000000000000000000000000000020000000000000001 --a 0 --b 0x1ee9
cargo run --release --example ghs_magic_spectrum
cargo run --release --example ghs_oakley_isogeny_census -- 300
cargo run --release --example ghs_relation_constants
```

Outputs in `data/`. `data/census-single-representative-l3-SUPERSEDED.txt` is the first run (l = 3, one twist representative per orbit); it is kept, not used.

Corrections to the first version of this report

1. The first cost model rows for m = 2, 3, 4 described magic numbers that cannot occur for n = 5 (finding 2); they and the “0.8 bits above rho” statement are withdrawn.

2. The census recorded one twist per j-orbit while saying it covered both (found in review of 85bcdcf0b); it now evaluates both and reports orbit and representative counts separately.
3. “Explicit characteristic-2 Velu isogeny not implemented” was wrong: `src/cryptanalysis/binary_velu.rs` has it. It was checked against Phi_3; a new test now also checks the codomain against the exact Phi_5 and Phi_7 (a rational order-l kernel gives j’ a root of Phi_l(X, j), and the group order is unchanged).

Findings

1. **Classical GHS.** Best magic number m = 5 (n = 5; l = 31 / 37), genus 2^(m-1) = 16 (type II). Other towers are far worse. MMT condition (8) holds. Neither curve is in the genus <= 4 range.

2. **The magic-number spectrum (new, exact).** m(b) = deg Ord_b + c (MMT Thm 5), where Ord_b divides x^n - 1 over **F_2** and c = 1 iff (x+1) does not divide Ord_b. By the normal basis theorem every divisor other than 1 occurs. For odd n the irreducible factors are the cyclotomic cosets of 2 mod n. `ghs_magic_spectrum` computes the spectrum and confirms it by enumerating every b != 0 in four small fields (N = 9, 10, 14, 15): observed = predicted in all four.

    - n = 5: spectrum {1, 5}. **Every curve not defined over the subfield has m = 5**; no isogeny can lower it. m = 1 is the subfield case, which the Oakley orders exclude by a size argument (not checked by computation here).
    - n = 37: {1, 37}. n = 31: seven values, smallest above 1 is 6. n = 155: 62 values. n = 185: 12 values.
    - So extended GHS by isogeny cannot reduce genus below 16 on the n = 5 tower. The question moves to Hess’s generalized covers (different Artin-Schreier functions), which this repository does not implement, and to the cost of HCDLP at genus 16.

3. **Isogeny census (both twists).** Phi_2 mod 2 = (X^2 - Y)(X - Y^2), so 2-isogenies are Frobenius/Verschiebung and identify conjugates. Moving by l in {3, 5, 7} over Frobenius orbits, 300 orbits per curve, 600 twist representatives:

    - Group 3: n = 5: all 600 have m = 5 (as finding 2 requires). n = 31: m in {26, 31}. n = 155: m in {126, 145, 146, 150, 151, 155}. Every observed value lies in the tower’s spectrum.
    - Group 4: l = 5, 7 leave the 3-isogeny-isolated orbit (the first run’s “no rational 3-isogeny” holds for l = 3 only). n = 5: all 600 have m = 5; n = 37: all 600 have m = 37; n = 185: m = 181 (48) or 185 (552). All lie in the tower spectra.
    - Walk is j-invariant level; no isogeny or kernel is built by the census.

4. **Cost model (exponent level, NOT a measurement).** Only m = 5 exists, so there is no walk cost:

curve	genus	Gaudry g!q^2 (constant 1)	Diem/Nagao q^(2-2/g), exponent only	rho
Group 3	16	2^106.3 (+29.8 bits vs rho)	2^58.1 (-18.4 bits, constant unknown)	2^76.5
Group 4	16	2^118.3 (+26.8 bits)	2^69.4 (-22.1 bits, constant unknown)	2^91.5

The two columns disagree in sign, so the exponent analysis does not decide it: the hidden constant of a genus-16 Diem-type relation search and its sparse linear algebra is the open cost question (Menezes-Teske, section 10). Menezes-Teske (ePrint 2004/235) record GF(q^5) as weak under plain GHS, so these curves sit in the regime the literature already flags. Units differ between columns and rho (field vs group operations); the section 5a band at 0, 0.1 and 1 GAE is not applied, so **no ordering of arms is claimed**.

Measured relation-phase constants (Gaudry route, genus 16)

`ghs_relation_constants` measures two inner loops in the same generic `F2mElement` arithmetic (unoptimized on both sides); output in `data/relation-constants.txt`. - **Smoothness probability.** A random monic degree-g polynomial over F_(2^8) splits into distinct linear factors with probability within 0.4% to 10% of 1/g! for g = 2..6 (ratio measured/1/g!: 0.996, 0.988, 0.980, 0.973, 0.904). The ratio falls with g because q = 256 is small next to g; at q = 2^31 the 1/g! value is the expected one but was not measured there (2^-44 is out of reach). The `x^q = x (mod u)` splitting test agrees with root finding on 5000 random degree-4 polynomials (0 disagreements). - **Cost per trial.** Degree-16 splitting test over F_(2^31): about 0.63 ms. Affine EC addition proxy (1 inversion, 2 mul, 1 square) over F_(2^155): about 7.4 microseconds. - **Result.** At genus 16 the relation phase alone needs about q/(1/16!) = 2^75.3 trials, 2^94.5 ns in total, against rho at 2^76.5 steps = 2^89.3 ns: **5.2 bits slower**, before the factor-base build, divisor arithmetic and linear algebra, so this is a lower bound for the plain Gaudry route. Optimized code would shift both sides by unmeasured, different factors, so the sign is a measurement of these two loops only, not of the best implementations. It says nothing about the Diem/Nagao route, which needs far fewer smoothness trials; that route is still unmeasured.

Distance remaining

Gaudry-type HCDLP: 27 to 30 bits above rho. Diem-type exponent: 18 to 22 bits below rho before constants. Neither is a measured end-to-end cost.

Verification run

`cargo test --release --lib` on this branch (with `phi_exact` and the l = 5..13 lookup): 4488 passed, 0 failed, 119 ignored (the repo's own `#[ignore]` markers), 419 s. The new `phi_exact_matches_tables_and_kronecker` test is among the passes.

Not done

- Hess generalized covers (direction 1 proper): needs function-field arithmetic. Magma and Sage are not installed; `pip install passagemath-standard` was tried and failed with "No space left on device" (about 3.3 GiB free; Sage needs several GiB more). The Menezes-Teske text could not be fetched (HTTP 403 on the PDF).
- Velu point-map transport at l >= 5 (the codomain is tested at l = 5, 7; the point map is tested at l = 3 only); the Diem/Nagao relation search, factor-base build and linear algebra (only the Gaudry smoothness loop is measured above).
- Conductor: the release download returned HTTP 403 through the sandbox proxy; `go install` through the module proxy worked, but `conductor up` needs Postgres or a Docker daemon, neither of which runs here, so `conductor check` was not run.
- No replay certificate, validator report or section 5a matched rho, so this is not eligible for the leaderboard or `benchmarks/`.

Figure

